TUTORIAL NO 1

Topics: Topics: *Prepositional & Predicate Logic from Unit 1 of the syllabus for the A.Y. 2025-26*

Q1. Define a proposition. Classify the following statements as propositions or not:

a) 5 + 3 = 10 c) x + 2 = 7

b) Close the door d) The sun rises in the east.

Q2. Construct the truth table for the compound proposition:

(i) $(p \wedge q) \rightarrow p$ (ii) $\neg(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \rightarrow r))$

Verify whether they are tautologies or not.

Q3. Using truth tables, verify the logical equivalence:

$p \rightarrow q \equiv \neg p \vee q$

Q4. Find the negation of the following statements:

a) All students are honest.

b) Some engineers are intelligent.

c) Every natural number is positive.

Q5. Write the converse, inverse, and contrapositive of the statement:

"If a number is divisible by 4, then it is even."

Q6. Simplify the following logical expression using laws of logic:

$(p \wedge q) \vee (p \wedge \neg q)$

Q7. Determine whether the following argument is valid:

If it rains, then the ground is wet.

It is raining.

∴ The ground is wet.

Name the rule of inference used.

Q8. Express the following statement symbolically:

"If I study or attend classes regularly, then I will pass the exam."

Let p: I study

q: I attend classes regularly

r: I pass the exam

Q9. Prove using logical laws:

$\neg(p \vee q) \equiv \neg p \wedge \neg q$

Q10. Determine whether the following statement is a tautology, contradiction, or contingency:

(i) $(p \vee \neg p) \rightarrow (q \vee \neg q) \rightarrow (r \vee \neg r)$ (ii) $\neg(p \vee (q \wedge r)) \leftrightarrow ((p \vee q) \wedge (p \rightarrow r))$